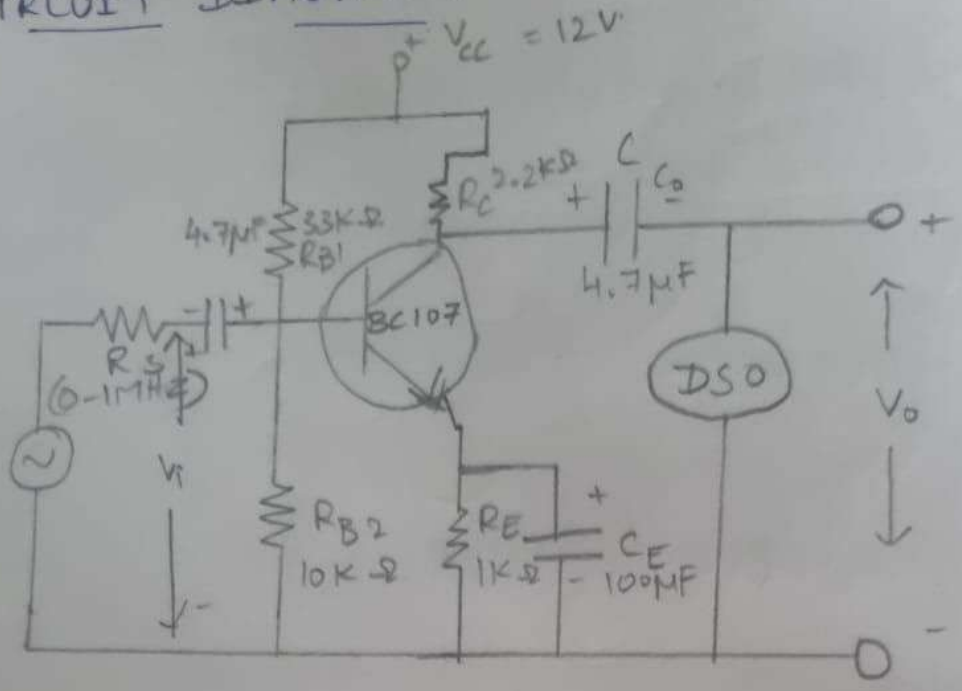
RESULT:-

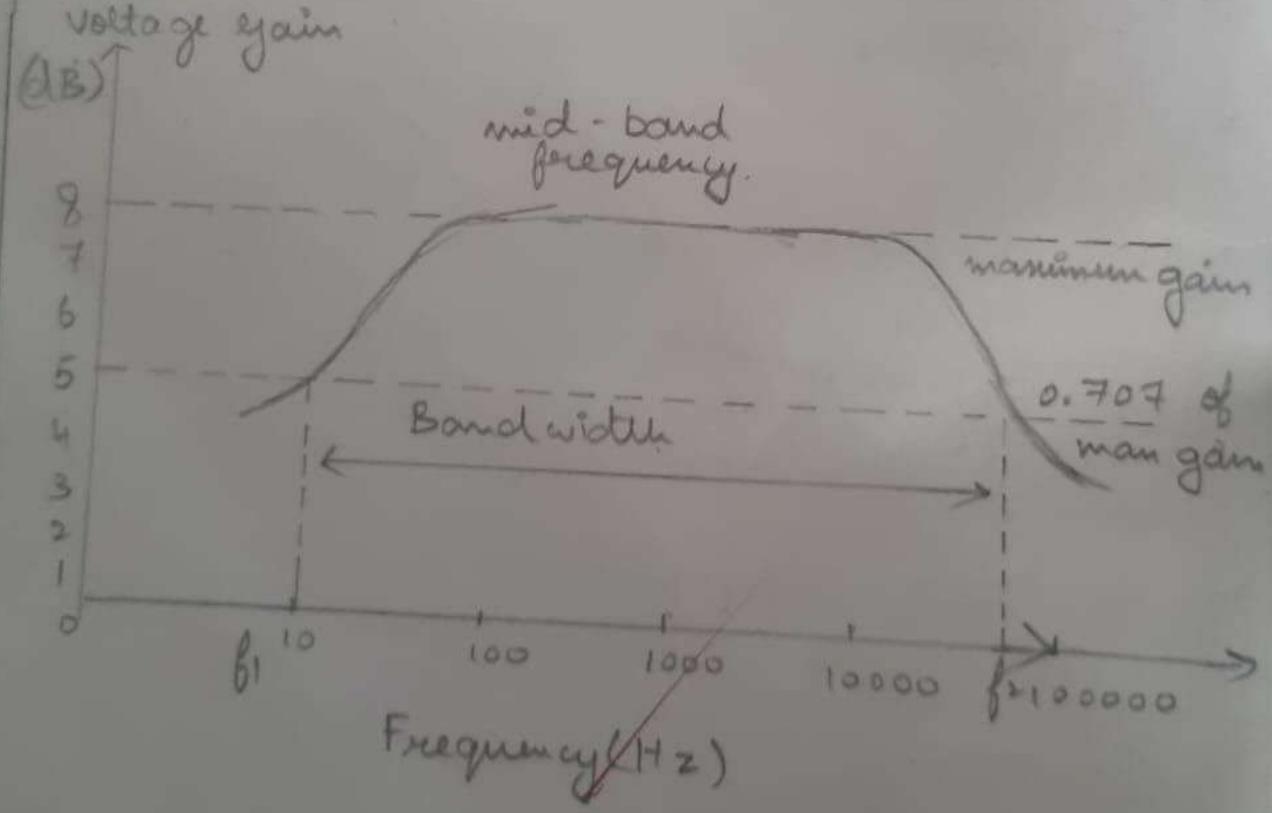
The common emitter amplifier was constructed and the following results were observed.

1. Maximum signal handling capacity
2. Q-point: I_{CQ}
   V_{CEQ}
3. Gain of the amplifier.
4. Bandwidth of the amplifier.
5. Gain-Bandwidth Product.

CIRCUIT DIAGRAM :-



MODEL GRAPH :-



Tabulation :-

S.NO	Frequency (HZ)	V_o(V)	V_o/V_{in}(V)	Gain = $20\log(V_o/V_{in})$ (dB)
1.	10	0.2	2	6.02 dB
2.	50	0.4	4	12.04
3.	100	0.6	6	15.56
4.	500	0.75	7.5	17.50
5.	1K	0.8	8	18.06
6.	5K	0.78	7.8	17.84
7.	10K	0.7	7	16.90
8.	50K	0.5	5	13.98
9.	100K	0.3	3	9.54

EXP NO: 01 FREQUENCY RESPONSE OF COMMON EMITTER AMPLIFIER.

AIM:-

To Design a single-stage common-emitter amplifier using a transistor, simulate the circuit in LT spice, analyze its frequency response, and calculate gain and bandwidth.

SOFTWARE REQUIRED:-

* LT spice simulator.

PROCEDURE:-

* Connect the circuit as per the circuit diagram using LT spice simulator.